a9lim

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PROFILE

I'm a Singaporean developer doing independent LLM interpretability research: activation steering, trait probing, and how much of a model's hidden state is recoverable from what it writes. All of my code is written by directing Claude and Codex end to end, and I dogfood everything I make: the agent tools below run my own daily workflow, and the introspection corpus comes from my own coding sessions. I'm looking for research roles and collaborations.

RESEARCH

Saklas: activation steering and probing library PyPI · Python

- Generates contrastive statement pairs and extracts steering vectors via contrastive PCA (Zou et al. 2023) to steer and monitor LLM hidden states, no fine-tuning; ships trait probes for affect, stance, and alignment.
- Python API, OpenAI- and Ollama-compatible server, and live webUI across 55 model architectures; AGPL-3.0 on PyPI with 1,000+ monthly downloads, CI, and llama.cpp GGUF interchange.

Introspection via Kaomoji Python · HuggingFace

- Asked whether the kaomoji a model opens its message with tracks its internal state: across five open-weight model families, the emotional category of the prompt is recoverable from the hidden state at up to 99% accuracy, and from the single kaomoji at 80% on the best models against an 11% chance rate.
- Found all five architectures recover the same affect geometry after Procrustes alignment, evidence toward the platonic representation hypothesis. Built llmoji, a CLI that hooks coding agents and publishes the corpus to HuggingFace; sibling studies in progress probe a residual-stream attractor basin and elapsed-time encoding.

EXPERIENCE

Independent Developer · a9l.im Feb 2026 – Present

- Built and maintain, solo, eight open-source interactive simulations plus the shared design system, component library, and Cloudflare Workers edge-SSR layer beneath them; roughly 29,000 lines of vanilla JavaScript.

SDDM Theme Maintainer · Catppuccin 2025 – Present

- Led the QtQuick rewrite of Catppuccin's SDDM display-manager theme (500+ stars): accent colors, user icons, vector backgrounds, and automated theme generation across all four flavors.

SELECTED PROJECTS

Bio-transformer Python

- Neuromorphic transformer from three brain subsystems that each compute something transformer-shaped: astrocytes as recurrent linear attention, the Marr–Albus cerebellum as the FFN, and laminar neocortex as the residual stream via predictive coding. Every neuron is a leaky integrate-and-fire spiker and every weight update is local (Oja, Hebbian, and climbing-fibre rules), with no backprop or surrogate gradient.

Agent Infrastructure Python · MCP

- Built the multi-agent tooling my own workflow runs on: gaslamp (two-way MCP bridge between Claude Code and Codex), gpqueue (cross-session GPU job queue with serial per-device slots), and dagwood (DAG-native task tracker with an MCP interface for agents).

Interactive Simulations · a9l.im JavaScript · WebGPU

- **Geon**: relativistic N-body simulator on WebGPU compute shaders, with 11 forces, Barnes-Hut scaling, and Kerr-Newman black holes. **Shoals**: options-trading sim with Heston, Merton, and Vasicek pricing and a 400+ scenario event engine. Six more cover resistive MHD, reactor neutronics, stochastic spatial epidemics, cellular metabolism, redistricting fairness, and a sixteen-text scripture reader with edge SSR.

Ogdoad Rust · Python

- Clifford algebras and quadratic-form classification over the number systems adjacent to Conway's combinatorial games: surreals recover the real Cl(p,q) classification with infinite and infinitesimal metric entries, while characteristic-2 legs like the nimbers swap the 8-fold periodicity for the Arf invariant and Brauer–Wall group. Verified Rust core with Python bindings.

EDUCATION

University of California, San Diego · B.S. Mathematics in 2.5 years · GPA 3.75 Sept 2023 – March 2026
Singapore American School · Summa Cum Laude · GPA 4.50 Class of 2023

SKILLS

Interpretability: activation steering, contrastive-PCA trait probing, and hidden-state analysis.
Agentic tooling: engineering by directing Claude Code and Codex; MCP servers; multi-agent orchestration.
Languages: Python (PyTorch, HuggingFace transformers, NumPy), JavaScript (Canvas, WebGL, WebGPU, GLSL, WGSL), Rust, Java, QtQuick and QML, LaTeX.
Other: Technical writing, vector graphics, soldering, Spanish (novice), and conlang construction.